

Project Documentation

Smart Lender: AI-Powered Loan Approval Prediction System

Repository Structure

```
Smart-Lender/  
|  
├─ app.py  
├─ test.py  
├─ requirements.txt  
├─ README.md  
├─ loan_prediction.xlsx  
├─ notebook/  
│   └─ Smart_Lender.ipynb  
├─ model/  
│   ├── loan_model.pkl  
│   └─ scaler.pkl  
├─ static/  
│   └─ style.css  
└─ templates/  
    ├── home.html  
    ├── input.html  
    └─ output.html
```

How to Run the Project

1. Install the required Python libraries using:

```
pip install -r requirements.txt
```

2. Open the project folder in Visual Studio Code or any Python IDE.
3. Ensure that the trained model files (loan_model.pkl and scaler.pkl) are present in the model folder.
4. Run the Flask application using the following command:

```
python app.py
```

5. After the Flask server starts successfully, open the local URL displayed in the terminal (for example, <http://127.0.0.1:5000>) in a web browser.
6. Enter the applicant details in the form and click Predict Loan Approval to view the prediction result.

References

- Dataset: Kaggle – Loan Prediction Dataset
 - Scikit-learn Documentation
 - Flask Documentation
 - Pandas Documentation
 - NumPy Documentation
 - Matplotlib Documentation
 - Seaborn Documentation
 - GitHub Documentation
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Conclusion

The Smart Lender project successfully demonstrates the use of Machine Learning for predicting loan approval decisions. By combining data preprocessing, model training, and a Flask-based web application, the system provides a fast and reliable method for evaluating loan applications. The project highlights the practical application of AI in the banking sector and serves as a foundation for future enhancements such as cloud deployment, user authentication, and real-time financial data integration.